USN | | | | | | | | | | [22EE41]

FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

STATIC AND DC ELECTRIC MACHINERY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Draw necessary circuit diagram wherever applicable.

ANSWER ALL QUESTIONS

		PART-A			**05x02=10**
1.	**a.**	Write a brief note on the power and distribution transformer.	1	**CO1**	**02**
	b.	Differentiate between Magnetic Flux Density & Magnetic Field Intensity	1	**CO1**	**02**
	c.	Enumerate various losses in an Electrical Machine.	2	**CO2**	**02**
	d.	Compare Dynamically induced emf & mutually induced emf.	2	**CO1**	**02**
	e.	Explain the factors governing the speed of a DC shunt motor and mention the methods of its speed control.	3	**CO4**	**02**
		PART-B			**06 X 05=30**
2.	**a.**	Draw and explain the detailed phasor diagram of a practical transformer supplying lagging power factor load.	2	**CO1**	**05**
		OR			
	b.	Show that the open delta (V-V) connection has a VA rating of 57.7 % of the rating of the normal Δ – Δ connection.	3	**CO5**	**05**
3.	**a.**	Draw and explain the speed/load characteristic of: **i.** DC Shunt **ii.** DC Series Motors and mention their applications.	2	**CO3**	**05**
		OR			
	b.	Describe in detail open circuit and short circuit on transformer. Determine the circuit constants, efficiency & regulation without loading the transformer.	3	**CO2**	**05**
4.	**a.**	Define the armature reaction and what all its effects on the machine, explain.	1	**CO3**	**05**
		OR			
	b.	Explain how two-phase supply can be obtained from a 3-phase supply using Scott connection draw the vector diagram.	2	**CO5**	**05**
5.	**a.**	With circuit diagram, explain Sumpner's test.	2	**CO5**	**05**
		OR			

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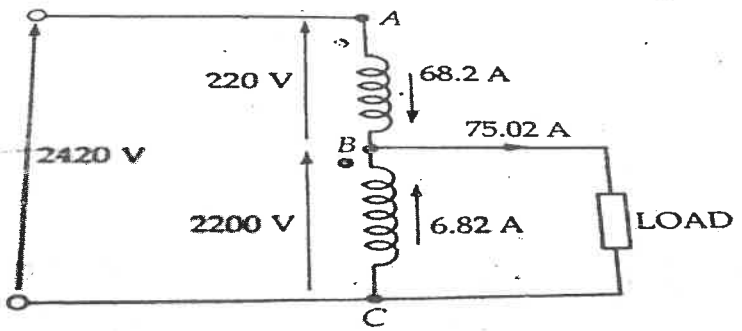
	b.	A 400 KVA, distribution transformer has full load iron loss of 2.5 kW and copper loss of 3.5 kW. During a day, its load cycle for 24 hours is, 6 hours 300 kW at 0.8 p.f. 10 hours 200 kW at 0.7 p.f. 4 hours 100 kW at 0.9 p.f. Determining it's all day efficiency	3	CO2	05
6.	a.	A 220 V . d.c. series motor is running at a speed of 800 r.p.m. and draws 100 A. Calculate at what speed the motor will run when developing half the torque . Total resistance of the armature and field is 0.1 Ohm.Assume that the magnetic circuit is unsaturated.	2	CO3	05
		OR			
	b.	A 120 V d.c. shunt motor having an armature circuit resistance of 0.2 ohm and field circuit resistance of 60 ohms, draws a line current of 40 A at full load .The brush voltage drop is 3 V and rated full load speed is 1800 r.p.m. Calculate : (a) the speed at half load ; (b) the speed at 125 per cent full load.	3	CO3	05
7.	a.	A 500 kVA transformer has an efficiency of 95% at full load and also at 60% of full load; both at UPF. **i.** Separate out the losses of the transformer. **ii.** Determine the efficiency of the transformer at 3/4th full load.	2	CO2	05
		OR			
	b.	List out the necessary and desirable conditions for parallel operation of two single phase transformer.	2	CO3	
		PART-C	**06 X 10=60**		
8.	a.	A single phase 250/500 V transformer gave the following results: Open circuit test 250 V , 1 A , 80 W on l.v. side Short circuit test 20 V, 12 A , 100 W on h.v. side Calculate the circuit constants and show them on an equivalent circuit.	2	CO3	10
		OR			
	b.	Explain the neat circuit and phasor diagrams the method of converting three phase supply to single supply. Show that load is balanced on both the sides.	2	CO5	10
9.	a.	Sketch and explain the various characteristics of D.C. Shunt Motor and mention its application.	3	CO4	10

		OR			
	b.	A 3 phase bank of three single-phase transformer are fed from 3-phase 33 kV (line to line) . It supplies a load of 6000 kVA at 11 kV (line to line) . Both supply and load are 3-wire. Calculate the voltage and kVA rating of the single -phase transformer foe all possible 3-phase transformer connection.	3	**CO3**	**10**
10.	**a.**	The Hopkinson test on two shunt machines gave the following results for full load: Line voltage, 250 V ; line current excluding field currents , 50 A ; motor armature current , 380 A ; field currents, 5 A and 4.2 A . Calculate the efficiency of each machine. Armature resistance of each machine 0.02 Ohm.	3	**CO5**	**10**
		OR			
	b.	A 200 V d.c. series motor runs at 1000 r.p.m. and takes 20 A. Combined resistance of armature and field is 0.4 ohms. Calculate the resistance to be inserted in series so as to reduce the speed to 800 r.p.m. Assume torque to vary as square of the speed and a linear magnetization curve.	3	**CO3**	**10**
11.	**a.**	Explain in detail with suitable diagram about Plugging & Regenerative Braking of DC shunt motor.	2	**CO5**	**10**
		OR			
	b.	A 2200/220 V transformer is rated at 15 kVA as a 2-winding transformer. It is connected as an autotransformer with low-voltage winding connected additively in series with high-voltage winding as shown in the figure below.  The autotransformer is excited from a 2420 V source . the autotransformer is loaded so that the rated currents of the windings are not exceeded.	3	**CO3**	**10**

		Find **i.** the current distribution in the windings. **ii.** kVA output. **iii.** kVA transferred conductively and inductively from input to output. **iv.** Saving in conductor material as compared to a two-winding transformer of the same VA rating.			
12.	**a.**	Develop the exact equivalent circuit diagram of a single-phase transformer. Derive the approximate and simplified equivalent circuit of the transformer.	2	**CO3**	**10**
		OR			
	b.	**i.** With the help of neat circuit diagram, Explain the WARD LEONARD method of speed control of D.C. Motor. **ii.** Write a short note on Swinburne's test for predetermining the efficiency of a DC machine.	2	**CO5**	**10**
13.	**a.**	With circuit diagram explain how efficiency of DC shunt machine can be obtained using regenerative test.	2	**CO5**	**10**
		OR			
	b.	A transformer has its maximum efficiency of 0.98 at 15 kVA at upf. Compare its all-day efficiencies for the following load cycles: **i.** Full load of 20 kVA 12 hours/day and no-load rest of the day. **ii.** Full load 4 hours/day and 0,4 full load rest of the day. Assume the load to operate on UPF all day.	3	**CO3**	**10**